one mole of liquid is converted into vapour state at its boiling point".

For example, heat of vaporisation of water can be represented as

$$H_2O(l) \xrightarrow{373K} H_2O(v) \quad \Delta H_{vap} = +40.626 \text{ kJ}$$

Molar heat of sublimation

Sublimation is a process when a solid changes directly into its vapour state without changing into liquid state. Molar heat of sublimation is defined as "the change in enthalpy when one mole of a solid is directly converted into the vapour state at its sublimation temperature". For example, the heat of sublimation of iodine is represented as

$$I_2(s) \xrightarrow{373K} I_2(v) \quad \Delta H_{sub} = +62.42 \text{ kJ}$$

Another example of sublimation process is solid CO_2 to gas at atmospheric pressure at very low temperatures.

Heat of transition

The heat of transition is defined as "The change in enthalpy when one mole of an element changes from one of its allotropic form to another. For example, the transition of diamond into graphite may be represented as

$$C_{(diamond)} \longrightarrow C_{(graphite)}$$

$$\Delta H_{trans} = +13.81 \text{ kJ}$$

Similarly the allotropic transitions in sulphur and phsphorous can be represented as follows,

$$S_{(monoclinic)} \longrightarrow S_{(rhombic)}$$

$$\Delta H_{trans} = -0.067 \text{ kJ}$$

$$P_{(white)} \longrightarrow P_{(red)}$$

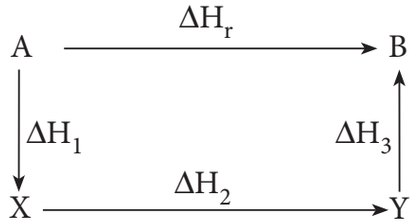
$$\Delta H_{trans} = -4.301 \text{ kJ}$$

7.8 Hess's law of constant heat summation

We have already seen that the heat changes in chemical reactions are equal to the difference in internal energy (ΔU) or heat content (ΔH) of the products and reactants, depending upon whether the reaction is studied at constant volume or constant pressure. Since ΔU and ΔH are functions of the state of the system, the heat evolved or absorbed in a given reaction depends only on the initial state and final state of the system and not on the path or the steps by which the change takes place.

This generalisation is known as **Hess's law** and stated as:

The enthalpy change of a reaction either at constant volume or constant pressure is the same whether it takes place in a single or multiple steps provided the initial and final states are same.



$$\Delta H_r = \Delta H_1 + \Delta H_2 + \Delta H_3$$

Application of Hess's Law: Hess's law can be applied to calculate enthalpies of reactions that are difficult to measure. For example, it is very difficult to measure the heat of combustion of graphite to give pure CO.

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